Home Network Security Redesign: From Flat Architecture to VLAN-Based Segmentation

Section 1: What Did I Do?

The foundation for this network redesign project began with the asset inventory assignment from CIDM 6341 that I took last semester. As a result of that inventory and my current studies for Network+, I realized that a redesign of my network was necessary. As I learned about VLANs and network segmentation, I began to recognize significant security gaps in my current flat network architecture. Both my Network+ course and this course emphasize that network segmentation serves as a critical defense-in-depth strategy, isolating different device types to limit the blast radius of a potential compromise.

This theoretical knowledge took on practical significance when I considered my home network's unique characteristics. Unlike a typical residential network, mine supports several self-hosted services running on a Proxmox virtualization server: a Vaultwarden password manager instance containing sensitive credentials, a NAS storing personal data and photo and video archives, and various other containerized applications accessible via Tailscale VPN. These services represent high-value assets that warranted stronger protections than a flat network could provide.

To systematically evaluate my network's security posture, I conducted a threat modeling exercise based on the principle of least privilege. I asked: what network access does each device type actually require? The analysis revealed concerning findings. IoT devices such as smart home equipment and streaming devices only need internet access and limited local network access for control via mobile apps, yet they currently have unrestricted access to all network resources. Guest devices should have internet-only access with no visibility into internal services. My homelab server requires access from trusted devices but should be isolated from potentially vulnerable IoT devices. Gaming consoles need high-bandwidth internet access but have no legitimate need to communicate with my password manager or NAS.

This gap analysis made clear that my network's flat topology violated fundamental security principles. A compromised IoT device would have network-level access to my Vaultwarden instance and personal files. With the security requirements established, I researched implementation approaches for network segmentation. My Network+ materials covered VLANs as the standard method for creating logical network separation, so I began evaluating my existing equipment's VLAN capabilities.

This research revealed a critical limitation of my setup. While my managed switch supports VLANs, the Deco router does not. Without VLAN-aware routing at the network edge, true segmentation wasn't possible with my current equipment. This discovery led to the next phase of research: evaluating equipment options that could provide proper VLAN routing and inter-VLAN firewall capabilities. I investigated dedicated firewall appliances, virtualized firewalls, and prosumer routers like Ubiquiti's equipment. Each option presented different trade-offs in terms of cost, complexity, and single points of failure.

Section 2: What Were the Results?

Network Users and Usage Patterns

The home network supports two primary users, a significant IoT device ecosystem, and occasional guest access. I represent the network's most demanding user from both technical complexity and bandwidth perspectives. Daily usage includes remote access to self-hosted homelab services via Tailscale VPN, large photo and video file transfers (often several gigabytes), multiple simultaneous device usage (laptop for coursework, tablet for media, phone for general use), PlayStation 5 gaming requiring low-latency connections, video streaming across multiple services, and Google Drive synchronization for backup redundancy.

My wife represents a standard residential internet user focused primarily on mobile devices. Her smartphone serves as her primary device for browsing, social media, and streaming. She occasionally uses a tablet and laptop, and her devices synchronize with iCloud for backups. An Apple Watch maintains connectivity for notifications and health data.

The network supports 13 connected IoT devices: two smart TVs, two Google Home Displays, two Nest Audio speakers, an Ecobee thermostat, six smart lights, and a Nest Doorbell. These devices generate consistent network traffic for cloud connectivity, firmware updates, and app-based control. Critically, none require access to internal network resources such as the homelab or NAS. Occasional visitors require temporary WiFi access that should be completely isolated from internal network resources.

Network Requirements Analysis

The current TP-Link Deco XE75 Pro mesh system with two nodes provides full coverage throughout my 1,700 square foot single-story home with no dead zones. The system delivers 2.4 GHz for extended range, 5 GHz for higher throughput, and 6 GHz for backhaul between nodes. Responsiveness is sufficient for gaming, streaming, and remote access to homelab services. The wireless mesh backhaul results in approximately 300 Mbps throughput loss compared to wired backhaul, but this still exceeds requirements given the 1 Gbps internet connection. The primary driver for redesign is security architecture, not performance deficiency.

Network availability requirements vary by service type. Critical services requiring 99% or higher uptime include internet connectivity for remote access to Vaultwarden, the Vaultwarden service itself, NAS availability for file access and workflow, and my laptop and homelab server reliability. High-priority services include smart home automation, streaming services, and mobile device connectivity. Lower-priority services that can tolerate temporary outages include guest WiFi and gaming console connectivity.

The current Vexus Fiber 1 Gbps symmetrical connection exceeds all usage requirements. Typical usage shows 4K streaming consuming approximately 25 Mbps per stream with up to two simultaneous streams totaling 50 Mbps. General browsing and IoT devices aggregate approximately 10 to 20 Mbps. Peak utilization rarely exceeds 200 to 300 Mbps, leaving substantial headroom. Internal network bandwidth proves more demanding due to photo and video workflow requirements involving multi-gigabyte transfers from laptop to NAS.

The home's wood-frame construction with drywall does not significantly impede wireless signal propagation. The requirement for a mesh system stems from infrastructure placement rather than home size. The fiber drop location in the office necessitates either wireless backhaul or ethernet cabling to extend connectivity. In the proposed redesign, ethernet backhaul would eliminate the 300 Mbps throughput penalty.

Proposed Network Design

The current Deco system, while providing excellent wireless performance, lacks VLAN support on wired connections. This necessitates transitioning to prosumer networking equipment. After evaluating several platforms including UniFi, TP-Link Omada, and Mikrotik, I selected Ubiquiti UniFi for its balance of capability and manageability.

The proposed equipment includes the UniFi Cloud Gateway Ultra ($129) to handle routing, VLAN configuration, and inter-VLAN firewall rules. Two UniFi U6 Lite Access Points ($99 each) provide WiFi 6 dual-band capability with per-SSID VLAN tagging. The existing TP-Link managed switch will be retained since it supports VLAN trunking. The UniFi Network Controller will be deployed as a container on the existing Proxmox server for self-hosted management.

The design implements ethernet backhaul to both access points, eliminating the wireless backhaul penalty. This requires running Cat6 cabling from the office to the living room AP location. Total cost approximates $427 to $627 depending on cabling approach, compared to $300 to $400 for consumer mesh systems.

VLAN Architecture

The network will be segmented into five logical VLANs serving distinct security and functional purposes:

The General VLAN (VLAN 10, 192.168.10.0/24) serves trusted user devices including laptops, smartphones, tablets, and other personal computing devices with full access to homelab services and internet connectivity.

The Homelab VLAN (VLAN 20, 192.168.20.0/24) hosts critical infrastructure including the Proxmox server, NAS, Vaultwarden, and other self-hosted services. Devices on this VLAN can initiate connections to the internet for updates, but strict firewall rules limit which VLANs can access homelab resources.

The Gaming and Entertainment VLAN (VLAN 30, 192.168.30.0/24) contains the PlayStation 5 and smart TVs. This separation prevents gaming traffic from interfering with critical services and provides the ability to implement quality of service rules. Smart TVs are placed here due to their entertainment function and higher bandwidth requirements.

The IoT VLAN (VLAN 40, 192.168.40.0/24) isolates potentially vulnerable smart home devices including the thermostat, doorbell, displays, speakers, and smart lights. This represents the highest security risk category. Devices on this VLAN have internet access for cloud connectivity but cannot initiate connections to other VLANs. The General VLAN can send commands to IoT devices for control purposes.

The Guest VLAN (VLAN 50, 192.168.50.0/24) provides internet-only access for visitor devices with complete isolation from all internal network resources.

Inter-VLAN Firewall Rules

The General VLAN can access the Homelab VLAN for file storage and password manager access, send control commands to the IoT VLAN for device management, and access the Gaming and Entertainment VLAN. The Homelab VLAN can initiate outbound internet connections but cannot access other VLANs unless explicitly permitted. The Gaming and Entertainment VLAN has internet access and can be accessed from the General VLAN but cannot initiate connections to Homelab or IoT VLANs. The IoT and Guest VLANs have internet access only with no access to any internal VLAN.

Each VLAN will have its own DHCP scope managed by the Cloud Gateway Ultra. Wireless SSIDs will be configured with VLAN tagging: a single SSID for trusted devices on the General VLAN, an IoT-specific SSID on the IoT VLAN, and a Guest SSID on the Guest VLAN. The Gaming and Entertainment VLAN will be accessed via wired connection for the PlayStation 5 and via the General SSID for smart TVs. The Homelab VLAN connects exclusively via wired ethernet.

Migration Strategy

The transition can be accomplished with minimal downtime through a phased approach: install and configure the Cloud Gateway Ultra alongside the existing Deco system, run ethernet cabling to AP locations, install U6 Lite access points with identical SSIDs and passwords, cutover routing from Deco to Cloud Gateway during a low-usage window, gradually configure VLANs and migrate devices starting with the IoT VLAN, and finally decommission Deco equipment once migration is validated. This phased approach allows testing at each step rather than a risky complete replacement.

Service Provider Selection and Service Level Agreement

Vexus Fiber provides the primary internet service with a 1 Gbps symmetrical fiber connection. In three years of service, only one outage has occurred. Vexus does not publish formal SLA guarantees for residential service, which operates on a best effort basis. However, empirical reliability has been exceptional with approximately 99.9% or higher uptime.

T-Mobile Home Internet provides backup connectivity through 5G cellular broadband. This service was acquired when T-Mobile purchased Vexus infrastructure in the region. While not as fast as the primary fiber connection, cellular backup provides redundancy since it uses different infrastructure with different failure modes. The UniFi Cloud Gateway Ultra supports WAN failover configuration for automatic switchover.

The network relies on several external cloud services. Both users depend on Gmail for email. I use Google Drive for backup with a 99.9% uptime SLA for paid tiers, while my wife uses iCloud with no published SLA. Streaming services are outage-tolerant and provide no formal consumer SLAs. Smart home cloud services from Google Home, Nest, and Ecobee publish no SLA guarantees, though most features maintain local control during internet outages.

Self-hosted services present different dependencies. Vaultwarden requires internet for remote access via Tailscale but functions normally on the local network during outages. The NAS maintains local network access at all times. The design philosophy prioritizes local functionality to reduce dependence on external service provider SLAs.

Network Technology Requirements

Gigabit ethernet provides the backbone for high-bandwidth, latency-sensitive devices including my laptop for file transfers and development work, the Proxmox homelab server, PlayStation 5 for gaming, and network infrastructure. All wired connections terminate at the TP-Link managed switch in the office, with additional cabling extending to AP mounting locations for ethernet backhaul.

Wireless connectivity serves mobile devices (phones, tablets, laptops when not docked), the entire IoT ecosystem (TVs, smart speakers, displays, thermostat, doorbell, lights), and guest devices. Requirements include dual-band capability with 2.4 GHz for IoT range and 5 GHz for device throughput, multiple SSID support for network segmentation, VLAN tagging per SSID for security isolation, seamless roaming between access points, and WPA3 security protocol.

Bluetooth serves short-range connectivity for headphones, Apple Watch, and a soundbar. Bluetooth operates as independent device-to-device connections without routing through the network gateway.

Mobile connectivity primarily occurs via WiFi when at home. Two smartphones, two tablets, and two laptops benefit from seamless roaming as users move through the home, making proper access point placement critical for uninterrupted calls, streaming, and app usage.

The 13-device IoT ecosystem represents the highest security risk. Diverse manufacturers create devices with varying security postures and cloud dependencies. The VLAN architecture isolates these devices on VLAN 40, preventing access to internal network resources. Firewall rules permit internet access for cloud connectivity and allow the General VLAN to send control commands, but IoT devices cannot initiate connections to other network segments.

Cloud services integration occurs at multiple layers. User-level cloud services are accessed via applications with the network providing transport. Smart home cloud services require persistent connections to manufacturer clouds, with outbound HTTPS permitted from the IoT VLAN while inbound connections are blocked. Self-hosted services use Tailscale VPN for encrypted remote access rather than port forwarding for better security through zero-trust architecture.

Section 3: What Did I Learn?

This assignment revealed the substantial complexity hidden within what appears to be a simple home network. Before this systematic analysis, I viewed my network through a functional lens: Does it provide WiFi coverage? Can I access my homelab? Are my speeds adequate? Breaking down the network using a structured methodology exposed layers of complexity I had not fully considered. Categorizing users by usage patterns, documenting every network dependency, analyzing availability requirements, and mapping security boundaries transformed my understanding from "it works" to "here's exactly how it works and where the vulnerabilities exist."

The most significant learning came from analyzing my IoT device ecosystem. I knew abstractly from my Security+ studies that IoT devices represent security risks through botnet attacks, default credentials, and lack of firmware updates. But this remained theoretical until I inventoried 13 IoT devices and recognized that a compromised smart light bulb currently has the same network access as my Vaultwarden password manager. This realization made the importance of network segmentation immediately tangible. VLANs shifted from "a feature enterprise networks use" to "a critical security control I need to implement."

Perhaps the most practical takeaway was learning to approach network design methodically rather than reactively. My previous approach was essentially "buy equipment that solves the immediate problem." This assignment forced a different approach: start with requirements including users, usage patterns, security needs, and availability expectations, then design a solution that addresses those requirements holistically. This structured methodology is directly applicable to my career transition into IT and cybersecurity roles. The same process applies whether designing a home network, a small business network, or enterprise infrastructure.

The assignment highlighted the value of my ongoing Network+ and Security+ certification studies. Concepts like VLANs, network segmentation, defense in depth, and least privilege access were familiar from textbooks, but applying them to my actual home network solidified that understanding. I now comprehend not just what a VLAN is, but why I would implement one, how it addresses specific security risks, and what trade-offs it involves. Conversely, the assignment exposed gaps in my consumer equipment. The Deco mesh system works excellently for its intended purpose, but it's designed for users who prioritize simplicity over granular control. Understanding this limitation helps me make better recommendations in future IT roles about when consumer equipment is appropriate versus when prosumer or enterprise gear is justified.

This assignment transformed the proposed network redesign from theoretical exercise to actionable implementation plan. I now have documented requirements, researched equipment options, cost estimates, and a migration strategy. While I may not implement the full UniFi solution immediately due to budget constraints, I have a clear roadmap for improving my network's security posture incrementally. More importantly, I have a framework for analyzing and designing networks that extends beyond this single project. The three-question methodology provides a repeatable structure for approaching technical challenges systematically rather than haphazardly.